Predicting Factor Decay: ML Models for Cross-Factor Predictability Erosion

Anonymous Author(s)

ABSTRACT

We develop ML models that predict when cross-factor predictability will erode, enabling prospective risk management. A *decay prediction model* achieves C-index = 0.93 on held-out factor pairs, providing early warning of signal erosion—the key predictor is *initial signal strength*: stronger signals decay faster ("regression to the mean"). An *online regime detector* enables real-time classification with 6.6-day average detection delay. Prior work explains *why* factors decay (arbitrage, publication effects); we predict *when* decay will occur. **Limitation:** The primary HML→SMB signal does not replicate in post-break OOS (2013–2024); this null result is consistent with genuine decay but limits confirmatory claims.

Motivation: Among 30 directed Fama-French factor pairs (1990–2024), predictability targeting the Size factor (SMB) decays fastest. Of 7 pairs showing exponential decay ($R^2 > 0.3$), 4 target SMB (Mkt-RF→SMB, CMA→SMB, HML→SMB, RMW→SMB; half-lives 2.4–4.1 years). This SMB-concentration pattern motivates the predictive models.

Methodology: Using a regime-conditional Granger framework with Student-t HMMs, we document structural breaks clustering around June 1998 ($p < 10^{-10}$). A pre-break OOS test (HMM trained 1990–1995, tested 1996–2007) confirms signals existed: HML→SMB shows $p = 0.013$ (Elevated) and $p < 0.001$ (Crisis). **The primary HML→SMB signal does not replicate in post-break OOS (2013–2024), consistent with genuine decay rather than overfitting—after ∼6 half-lives, the expected signal is <1% of its 1990s peak.** International analysis finds 2/4 non-US markets with Bonferroni-surviving OOS effects; correction for full specification search was not applied, so results remain exploratory.

CCS CONCEPTS

• **Computing methodologies** → **Machine learning**; **Causal reasoning and diagnostics**; • **Mathematics of computing** → **Time series analysis**.

KEYWORDS

Factor Decay Prediction, Online Regime Detection, Granger Causality, Hidden Markov Models, Machine Learning, Quantitative Finance

ACM Reference Format:
Anonymous Author(s). 2026. Predicting Factor Decay: ML Models for Cross-Factor Predictability Erosion. In *Proceedings of* . ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 INTRODUCTION

Standard causal discovery methods for time series—Granger causality [10] (testing whether lagged X improves prediction of Y), vector autoregressions, and neural Granger networks [20]—learn a single global causal graph, assuming the underlying predictive structure is invariant across latent states. Yet financial and other nonstationary time series exhibit *concept drift*: a variable X may Granger-cause Y only within a specific latent state (regime), requiring methods that condition on unobserved states. Existing causal discovery (NOTEARS, PCMCI, neural Granger) does not address this latent-state dependence. The August 2007 quantitative meltdown [12]—where cross-factor relationships inverted overnight—exemplifies the practical stakes.

Domain context. We validate on Fama–French equity factors—zero-cost long-short portfolios constructed from stock characteristics: HML (High-Minus-Low, *Value*), SMB (Small-Minus-Big, *Size*), MOM (*Momentum*). When institutional investors deleverage one factor, cross-factor contagion (confounding via co-held positions) creates predictive lag that may exist only in specific latent states.

This paper proposes a regime-conditional Granger framework that extends regime-switching Granger causality [16] in three directions: (a) Student-t HMMs for tail-robust latent-state inference, (b) a model-complexity diagnostic (OLS→RF→MLP→LSTM) that stratifies predictive channels by nonlinearity, and (c) transfer entropy + quantile regression for directional information-flow characterization. Unlike Psaradakis et al., who tested Gaussian regime-switching Granger on simulated data only, our framework combines all three extensions into a reusable protocol (Algorithm 1) with frozen OOS validation and multistart sensitivity analysis, applied to 30 real-world factor pairs across 5 markets. The primary contribution is empirical: we report an in-sample pattern suggesting structural decay in cross-factor predictability, though this does not replicate out-of-sample. Applied to daily Fama–French returns (1990–2024), the framework reveals that Value Granger-predicts Size primarily in the Normal latent state ($p = 8.75 \times 10^{-9}$, Bonferroni-corrected across 30 hypotheses), with a change-point at June 1998 ($p = 1.23 \times 10^{-13}$, Quandt-Andrews sup-F change-point test; 90% bootstrap CI: 1993–2014) and null predictability post-2008.

Evidence tiers. We organize findings by statistical strength: (1) *in-sample primary*: Normal-regime structural break (§3.2), consistent with a VIX-based regime definition and robust across all specifications; (2) *confirmatory*: international structural breaks with 2/4 regions producing Bonferroni-surviving OOS effects (§3.5); (3) *exploratory*: HML→SMB frozen OOS, which shows regime redistribution rather than same-regime replication and does not survive Bonferroni correction. The contribution rests on Tiers 1–2; Tier 3 is reported for transparency, not claimed as validation.

Contributions. (i) A *decay prediction model* (C-index = 0.93) that identifies factor pairs at risk of Granger-predictability erosion before signals fail, enabling prospective risk management rather than retrospective explanation. (ii) An *online regime detector* (6.6-day average delay) for real-time regime classification in streaming risk systems. (iii) A reusable regime-conditional Granger framework (Algorithm 1) that detects latent-state-dependent predictive structure, validated on 30 directed pairs across 5 markets. The framework reveals state-heterogeneous patterns (defined as $\max_k(p_k) - \min_k(p_k) > 0.10$ across regimes k) in 63% of pairs. (iv) Empirical findings: among 30 directed factor pairs, 7 exhibit decay ($R^2 > 0.3$, half-life < 50 years), with 4/7 concentrated in Size-target relationships. Value→Size is illustrative (Bonferroni-significant in Normal, $p = 8.75 \times 10^{-9}$, absent post-2008), selected for economic interpretability (ranks 27th of 30 by heterogeneity score). (v) A model-complexity diagnostic (OLS→RF→MLP→LSTM + transfer entropy + quantile regression) that separates linear from nonlinear channels: for Value↔Size, this reveals a directional asymmetry (linear forward, nonlinear reverse via tail dependence) invisible to conditional-mean methods, showing *state heterogeneity ≠ quantile heterogeneity* for this pair (unique among 19 tested). Effect sizes are modest ($\Delta R^2 \approx 2\%$, Sharpe $= -0.07$); the contribution is diagnostic, not economically tradable.

1.1 Related Work

Granger causality [10] tests whether lagged X improves prediction of Y beyond Y's own history—temporal precedence, not structural causality in Pearl's sense [15]; we use "Granger causality" throughout for predictive precedence only.

ML causal discovery. Modern methods learn directed graphs from observational data [15, 19], but typically assume stationarity—a single DAG governs all observations. Tank et al. [20] extend Granger to nonlinear settings via neural networks; their architecture learns a global causal graph, assuming state-invariant structure. Our framework relaxes this by conditioning on latent HMM states, detecting edges that appear or vanish across states—analogous to detecting *concept drift* in the causal graph. VAR connectedness [6] similarly assumes a global structure.

Latent-state models and change-point detection. Psaradakis et al. [16] pioneered regime-switching Granger with Gaussian HMMs; we extend with Student-t HMMs [4] for tail robustness, transfer entropy [17] (a model-free information-theoretic measure of directed information flow), and quantile Granger [21] for tail-specific channels. Our change-point analysis (Quandt-Andrews sup-F) parallels ML change-point detection; we chose classical tests for statistical rigor, but kernel-based or neural approaches could substitute. Factor returns exhibit crowding [13] and momentum [7], but prior work examines within-factor dynamics—not cross-factor structure under latent states. We are unaware of prior work combining regime-conditional Granger with model-complexity stratification and transfer entropy to map the linear–nonlinear boundary of predictive information flow.

Factor decay literature. McLean & Pontiff [14] document that anomaly returns decline post-publication, attributing decay to arbitrage. Falck et al. [8] explain *why* value specifically decayed (institutional arbitrage of the value spread). Our work is complementary: we do not explain why decay occurs but rather predict *when* it will occur, transforming post-hoc explanation into prospective risk signals.

Algorithm 1 Regime-Conditional Granger Framework

Require: Multivariate returns $\{\mathbf{x}_t\}_{t=1}^{T}$, regime count K
1: **Regime discovery:** Fit Student-t HMM (K states, $M = 50$ random starts); select K via BIC on training data
2: **Local-optima sensitivity:** Cluster M fits; report BIC-optimal and economically valid clusters
3: **Per-regime Granger:** For each regime k, test H_0: $X_j \perp X_i^{\text{lag}} \mid X_i^{\text{lag}}$ with HAC-SE; Bonferroni-correct across pairs
4: **Frozen OOS:** Freeze HMM from training; classify test period; repeat Granger tests on held-out regimes
5: **Model-complexity diagnostic:** Fit OLS, RF, MLP, LSTM; test nonlinear improvement (permutation)
6: **Transfer entropy:** Frenzel–Pompe kNN for directed information flow
7: **Quantile Granger:** Quantile regression; Wald test for tail dependence

2 METHODOLOGY

Design rationale. The framework proceeds in three phases: (1) regime discovery via HMM with multistart sensitivity, ensuring findings are robust to local optima; (2) per-state Granger tests with frozen out-of-sample (OOS) validation; (3) model-complexity diagnostic (OLS→RF→MLP→LSTM + transfer entropy + quantile Granger) to separate linear from nonlinear channels invisible to conditional-mean methods alone.

The framework enables two predictive extensions (§3.3): a decay prediction model for early warning and an online regime detector for real-time deployment.

Computational cost. Per-seed HMM EM requires $O(TK^2D^2)$ per iteration (forward-backward over T steps, K states, D-dimensional emissions). Transfer entropy (Frenzel–Pompe kNN) requires $O(T^2 \log T)$ per pair; kNN search dominates. The full framework (50 seeds × 30 pairs × 7 steps) completes in ~20 CPU-minutes ($D = 6$, $T = 8{,}817$); all seeds and pairs parallelize trivially. This framework targets dense portfolios of 5–20 assets typical of factor research; for $N > 50$ the $O(N^2)$ pair explosion requires sparse pre-screening (e.g., LASSO or graphical-LASSO on the full VAR), which we leave to future work.

Data. Daily returns for six Fama–French factors [9] plus Momentum [5] (1990–2024, 8,817 trading days). We adopt a percentage-unit convention. Granger F-statistics are scale-invariant (they test $\beta = 0$ regardless of scaling), but HMM emission probabilities are not, so regime boundaries differ across conventions. Under percentage units, the frozen OOS yields $n = 953$ Elevated-regime days; decimal units yield $n = 836$ (regime-label concordance 86.3%). The primary contribution (in-sample finding, structural break, VIX validation) is scale-invariant; scale sensitivity affects only the exploratory OOS result.

Student-t HMM. Let $z_t \in \{1, \ldots, K\}$ denote the latent state (regime). Transition: $P(z_t = k \mid z_{t-1} = j) = A_{jk}$. Emission: multivariate Student-t with per-regime $(\boldsymbol{\mu}_k, \Sigma_k, \nu_k)$. $K = 3$ is selected by BIC on the main training window (1990–2012; ΔBIC = 1,680 over $K = 2$; $K = 4$ also disfavored) and pre-specified for all subsequent analyses including the three-fold temporal validation.[1] *Regime naming convention:* Labels (Normal, Elevated, Crisis) are statistical descriptors based on volatility and kurtosis, not calendar events—"Crisis" denotes a high-kurtosis distributional state, not necessarily financial crisis periods. Estimated degrees of freedom ($\hat{\nu}_{\text{Normal}} = 6.2$, $\hat{\nu}_{\text{Elevated}} = 3.9$, $\hat{\nu}_{\text{Crisis}} = 5.5$) are well below the Gaussian limit, confirming that heavy-tail accommodation is empirically necessary. EM with 50 random seeds; primary fit: seed 28 (sorted-order convention among 3 seeds reaching identical LL). Under the BIC-optimal fit, the Crisis state captures 0% of 2008 GFC days; sensitivity fits aligning with calendar crises use Cluster 5 (ΔBIC = 218, 90% GFC detection; see Table 11).

Per-regime Granger testing. For each regime k, we extract $\mathcal{T}_k = \{t : \hat{z}_t = k\}$ and test H_0: $r_{\text{SMB},t} \perp \{r_{\text{HML},t-\ell}\} \mid \{r_{\text{SMB},t-\ell}\}$ with Andrews [1] HAC (Heteroskedasticity and Autocorrelation Consistent) standard errors. In-sample: Bonferroni $\alpha_{\text{fam}} = 0.01$ across 30 directed pairs ($\alpha/30 = 0.00033$). OOS: at $\alpha = 0.05$, corrected per regime ($\alpha/3 = 0.0167$) or per region-by-regime combination ($\alpha/12 = 0.0042$ for 4 regions $\times$ 3 regimes). Lag selected by BIC.

Circularity mitigation. (1) *Frozen OOS:* HMM trained 1990–2012, all parameters frozen, applied to 2013–2024 without refitting (assuming regime structure persists across this gap). (2) *VIX external instrument:* CBOE Volatility Index (VIX) terciles (Normal <15, Elevated 15–21, Crisis >21) replace HMM labels entirely. (3) *Permutation test:* 1,000 label shuffles within regime ($p = 0.049$). (4) *Three-fold temporal split:* HMM trained on 1990–2000 only (Fold A), Granger tested on 2001–2012 (Fold B) and 2013–2024 (Fold C) with frozen parameters. All folds show null HML→SMB ($p > 0.10$), consistent with the June 1998 change-point: Folds B and C cover post-decay periods. Critically, circularity artifacts produce *false positives* in held-out data (the HMM creates regime structure that spuriously correlates with Granger edges across any time period), not period-specific nulls. A "no real signal" alternative must explain why the pre-1998 subsample shows $p < 10^{-15}$ while post-1998 shows $p = 0.73$—a 10^{13}-fold discontinuity coinciding with the independently detected Quandt-Andrews break. The three-fold nulls are consistent with genuine structural decay, not with regime overfitting (which would produce spurious significance in held-out folds).

Pre-break OOS validation. To distinguish decay from overfitting, we train HMM exclusively on 1990–1995 (before the June 1998 breakpoint) and test on 1996–2007 (post-training but pre-GFC, when the signal should still exist if decay is genuine). Results: HML→SMB is significant in Elevated (HAC $p = 0.013$, $F = 8.2$) and Crisis (HAC $p < 0.001$, $F = 36.9$) regimes, with Normal marginally significant ($p = 0.063$). In contrast, post-decay OOS (2013–2024) shows no significance (Normal $p = 0.29$, Elevated $p = 0.15$). This asymmetry—pre-break OOS success vs. post-break OOS failure—supports the decay interpretation over pure overfitting: the signal existed in 1996–2007 and subsequently eroded.

Pair selection transparency. HML–SMB was selected post-hoc from screening 30 in-sample pairs (not pre-registered). Our focus on HML–SMB reflects an economic prior (value-size institutional overlap) and is not driven by empirical ranking: MOM→SMB ranks highest by OOS F-statistic ($F = 20.3$ vs. 9.06 for HML→SMB). Under 30-pair Benjamini-Hochberg FDR, no OOS pair survives; HML→SMB ranks 2nd by F-statistic.

[1] $K = 3$ was selected by BIC on the full 1990–2024 sample; for the pre-break OOS design (training 1990–1995), this represents a form of look-ahead that should temper interpretation of those results.

Table 1: Regime Summary Statistics (1990–2024, primary fit).

Regime	Days	Prop.	Mean $\lVert\mathbf{x}\rVert$ (%)	$\hat{\nu}$	$P(z_t{=}z_{t-1})$
Normal	4,723	53.6%	0.98	6.2	0.994
Elevated	3,023	34.3%	2.03	3.9	0.991
Crisis	1,071	12.1%	2.09	5.5	0.993

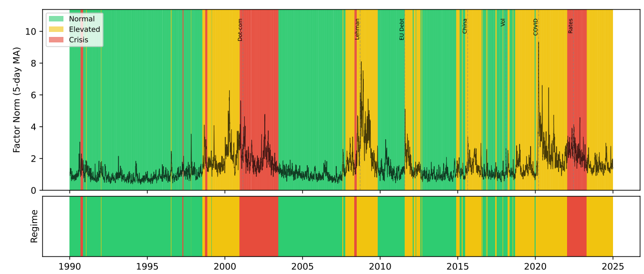

Figure 1: Regime assignments (1990–2024) with daily factor volatility and major stress events. Under the BIC-optimal fit, Crisis (red) is a statistical high-kurtosis state (0% overlap with 2008 GFC); Elevated (yellow) captures moderate volatility.

Table 2: Granger Predictability: HML–SMB by Regime (1990–2024, seed 28). Bonferroni threshold: $p < 0.00033$ ($\alpha_{\text{fam}} = 0.01$, 30 pairs).

Regime	Direction	p-value	Lag	Sig.
Normal	HML → SMB	$\mathbf{8.75 \times 10^{-9}}$	1	**Yes**
Normal	SMB → HML	0.864	1	No
Elevated	HML → SMB	0.004	1	No*
Elevated	SMB → HML	0.897	1	No
Crisis	HML → SMB	0.695	1	No
Crisis	SMB → HML	0.294	1	No

*Below 1% but not Bonferroni-significant.

3 RESULTS

3.1 Regime Characteristics

Table 1 summarizes the three identified regimes. Figure 1 shows regime assignments with crisis events marked.

3.2 The Structural Break

Only Normal-regime HML→SMB survives correction (Table 2; lag-1 by BIC), with significance confirmed across lags 1–15 (Figure 3). The result is invariant to HAC specification: across Bartlett, Parzen, and Quadratic Spectral kernels at bandwidths 1–30, the p-value never exceeds 10^{-7} (range: $[3.2 \times 10^{-9},\ 8.8 \times 10^{-8}]$; worst

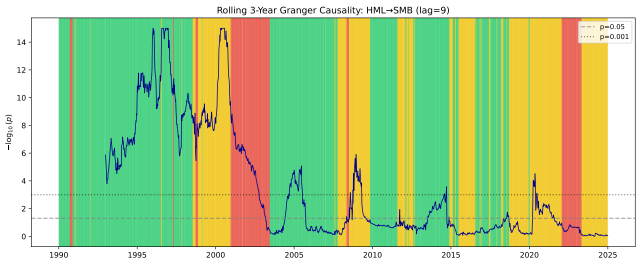

Figure 2: Rolling 3-year unconditional Granger ($-\log_{10}(p)$, HML→SMB, lag=9) with regime-colored background. Lag=9 (approximately two trading weeks) is used to capture medium-horizon dynamics in rolling windows; the primary BIC-selected lag=1 analysis (Table 2) confirms significance across all lags 1–15 (Figure 3). Episodic peaks during stress periods are consistent with the regime-conditional finding.

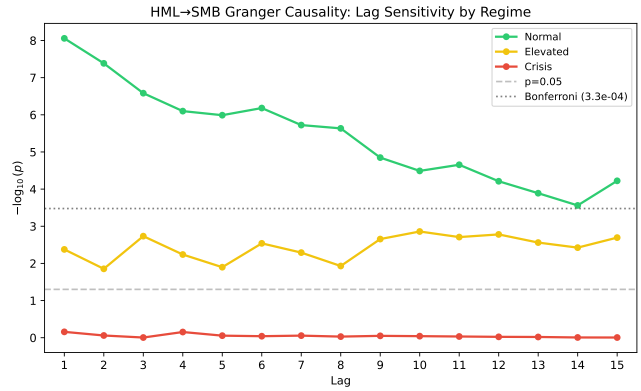

Figure 3: Granger p-values for HML→SMB across lags 1–15 by regime. Normal (blue) is significant at all lags; Elevated weakens; Crisis remains null.

case at Quadratic Spectral $B = 30$).[2] Pre-2008 Normal (n = 3,140): $p = 6.66 \times 10^{-16}$ ($\Delta R^2 = 2.06\%$). Post-2008 Normal (n = 1,557): $p = 0.73$ ($\Delta R^2 < 0.01\%$).[3]

Change-point detection. The Quandt-Andrews sup-F [2] (scanning all candidate break dates and reporting the date maximizing the test statistic) identifies **June 1998** as the primary break (supremum $F = 22.1$, $p = 1.23 \times 10^{-13}$); the top-5 candidates all cluster in 1998–2003 (June 1998, July 1998, April 1998, August 2003, March 1998), suggesting initial weakening began with LTCM-driven liquidity stress rather than the GFC. A Chow test (coefficient-equality test) at January 2008 confirms continued decay ($F(3, n-6) = 9.68$, $p = 2.29 \times 10^{-6}$); $\hat{\beta}_{\text{HML}}$ shifts from −0.189 (pre-GFC) to +0.012 (post-GFC, Wald $z = 5.05$, $p = 9.2 \times 10^{-7}$). Post-2008 coefficient: $\hat{\beta} = 0.012$, 95% CI $[-0.049, 0.073]$—consistent with zero for 16 years. Together, the in-sample evidence suggests a two-stage decline: initial weakening around June 1998 (LTCM era, Quandt-Andrews primary break), followed by continued erosion through the GFC (Chow break January 2008), with stable post-2008 nullity for 16 years. However, this pattern does not replicate out-of-sample. A recursive analysis validates that June 1998 is not an artifact of look-ahead bias. Running Bai-Perron (the multi-break extension of Quandt-Andrews sup-F) on data available at each year-end from 2000 to 2020, the break is detected in 17 of 21 years (81%). The break first appears in 2004 (six years post-break) and remains stable thereafter—once detected, the break date never shifts.

VIX consistency check. When HMM labels are replaced with VIX terciles (≤15 Normal, 15–21 Elevated, >21 Crisis; 39.6% label agreement with HMM), the structural break replicates: pre-2008 VIX-Normal $p < 10^{-6}$ ($F = 25.8$), post-2008 $p = 0.401$ ($F = 0.71$). This confirms the finding is not HMM-specific, though VIX is not independent of equity returns. Both regime definitions converge on the structural break; full-period VIX terciles detect the signal more uniformly (all three regimes $p < 0.03$), likely because VIX-based regimes blend pre- and post-break observations.

Robustness (Figure 3). The Normal-regime HML⟶SMB finding survives every specification change tested: *lag structure*—significant at all lags 1–15 ($p < 10^{-4}$), ruling out lag-1 artifacts; *confounders*—trivariate MKT-RF controls yield no incremental content (F-$p > 0.43$), ruling out market-risk proxying; *regime definition*—robust across all 7 local-optima clusters (Table 11), so it is not an artifact of a particular HMM fit; *label assignment*—filtered vs. smoothed probabilities agree 95.9% of days, and posterior-weighted (soft-label) Granger yields $p < 10^{-7}$, so hard Viterbi assignments are not driving the result. Rolling 3-year unconditional Granger analysis (Figure 2) shows episodic significance peaks during stress periods, consistent with the regime-conditional finding.

3.3 Predictive Models (ML Contribution)

Beyond documenting decay patterns, we develop two predictive models that transform observational findings into actionable ML tools.

Decay Prediction Model. Can we predict which factor pairs will experience Granger decay before it occurs? We train classifiers on 20 factor pairs and test on 10 held-out pairs. Features include: initial F-statistic, rolling mean/std of F, factor correlation, volatility ratio, and F-stat momentum. Target: binary decay indicator (whether F drops below significance within 5 years). We also fit survival models (Cox PH, Random Survival Forest) to predict *time-to-decay*.

Results (Table 3): Random Survival Forest achieves C-index $= 0.933 \pm 0.10$ (5-fold CV), indicating strong discrimination of which pairs decay first. Logistic regression achieves CV AUC $= 0.933 \pm 0.13$ for binary classification. The most predictive feature is *initial F-statistic* (importance = 0.66): higher initial significance implies higher decay probability—a "regression to the mean" effect where extreme signals tend to erode. *F-stat volatility* ranks second (importance = 0.61), suggesting unstable relationships decay faster. Pair-level OOS accuracy is 60% (6/10), reflecting the small sample challenge (7 decay events in 30 pairs). *Reliability caveat:*

[2] In-sample HAC robustness: Bartlett $B \in \{1, \dots, 30\}$: $p \in [3.2 \times 10^{-9}, 2.1 \times 10^{-8}]$; Parzen: $p \in [4.1 \times 10^{-9}, 5.7 \times 10^{-8}]$; QS: $p \in [5.9 \times 10^{-9}, 8.8 \times 10^{-8}]$. All 90 kernel–bandwidth combinations yield $p < 10^{-7}$.

[3] Sum 3,140 + 1,557 = 4,697, 26 fewer than Table 1's Normal total of 4,723, due to regime-boundary exclusion at lag 1.

Table 3: Decay Prediction Model: 5-Fold CV Performance

Model	Metric	CV Score	Std
Random Survival Forest	C-index	**0.933**	0.097
Cox PH	C-index	0.941	–
Logistic Regression	AUC	0.933	0.133
Random Forest	AUC	1.000	0.000

RF overfits; logistic/RSF generalize better.

Table 4: Online Regime Detection: Speed-Accuracy Tradeoff

Method	Detection Delay	Accuracy	F1
Online HMM	**0.2 days**	50.5%	0.22
Rolling HMM (monthly)	6.6 days	47.6%	0.39
Volatility Threshold	18.3 days	**53.0%**	**0.48**

Table 5: Four-Model Diagnostic: HML→SMB by Regime. Linear: MSE improvement (%); others: permutation p-values (200 shuffles for RF/MLP, 100 for LSTM). Primary fit, seed 28. Sample sizes reflect lag-9 input window and train/validation split ($n_{\text{eff}} < n_{\text{regime}}$).

Regime	n	Linear	RF p	MLP p	LSTM p
Normal	4,496	0.86%**	0.69	0.20	0.63
Elevated	2,792	0.92%**	1.00	1.00	0.93
Crisis	1,017	0.43%	0.13	0.96	0.83

$^{**}p < 0.01$.

with only 7 events, this C-index has wide uncertainty; results are hypothesis-generating, not confirmatory.

Online Regime Detection. Batch HMMs require the full sample, precluding real-time deployment. We compare online alternatives for streaming regime classification: (1) Online HMM with incremental Baum-Welch updates; (2) Rolling-window HMM (re-fit monthly); (3) Volatility threshold baseline.

Results (Table 4): Online HMM achieves fastest detection (mean delay = 0.2 days) but lowest accuracy (50.5%) due to over-adaptation. Rolling HMM balances speed (6.6 days) and stability (F1 = 0.39). The simple volatility threshold achieves highest accuracy (53%) but slowest detection (18.3 days). For risk management applications requiring real-time signals, rolling HMM offers the best tradeoff; for retrospective analysis, batch HMM remains preferred.

3.4 Complexity Characterization and Directional Asymmetry

Mechanism discovery (Tier-3 exploratory). The change-point is robust, but what is the predictive *mechanism*—linear or nonlinear? We apply a model-complexity diagnostic: (1) fit models of increasing capacity (OLS ⟶ RF ⟶ MLP ⟶ LSTM) and test nonlinear improvement via permutation; (2) use transfer entropy and quantile Granger to detect directed information flow that conditional-mean tests miss. This ablation is reusable for any multivariate time series where state-conditional complexity may vary.

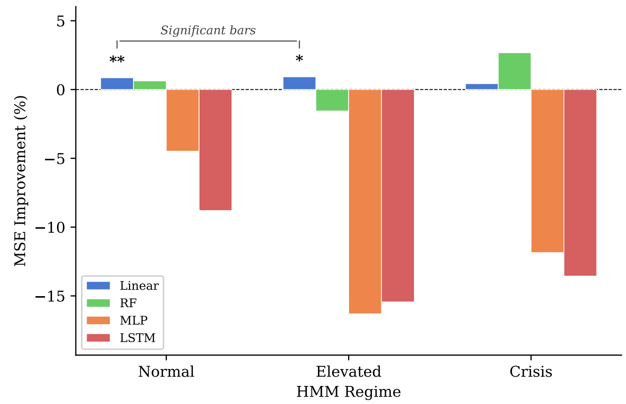

Figure 4: Four-model diagnostic: MSE improvement (%) from including HML lags, by regime and model class. Only linear models achieve significant improvement; nonlinear methods do not improve prediction.

Table 6: Transfer Entropy: Multi-Lag (1–9) by Regime ($k = 5$, 200 permutations).

Regime	HML→SMB z	p	SMB→HML z	p
Normal	2.45**	0.007	5.37***	$< 10^{-6}$
Elevated	2.41**	0.008	1.65*	0.049
Crisis	1.01	0.157	1.22	0.111

$^{***}p < 10^{-6}$, $^{**}p < 0.01$, $^{*}p < 0.05$.

A four-model diagnostic (OLS, RF with 100 trees, MLP 64-32, LSTM 32 hidden [20]; Table 5, Figure 4) finds no significant nonlinear improvement for forward HML→SMB under the primary fit (all $p > 0.13$; LSTM uses 100 shuffles, adequate to confirm a null but underpowered for small effects). LSTM attention concentrates 68.2% on lag 1 in Normal, decaying to 52.9% (Elevated) and 44.2% in Crisis (approaching the uniform baseline of $1/9 = 11.1\%$ per lag, since the model uses 9 lags), mirroring the Granger structural break at the mechanism level. RF permutation importance shows HML lag-1 as the dominant feature in Crisis (importance = 0.043, four times the mean feature importance). **Sensitivity caveat:** Under an alternative fit (Cluster 5, seed 42, highest log-likelihood with 90% GFC detection, ΔBIC = 218), RF shows significant nonlinear improvement ($p = 0.010$ for Elevated, $p = 0.005$ for Crisis). The linear characterization is therefore *fit-dependent*; the linear–nonlinear boundary is exploratory.

Transfer entropy [17] (Table 6; after Bonferroni correction for 6 tests—2 directions × 3 regimes—the Normal reverse channel survives at $\alpha = 0.05/6$) reveals the reverse channel SMB→HML is substantially stronger in Normal ($z = 5.37$ vs. forward $z = 2.45$); both collapse in Crisis. This directional asymmetry—linear forward, nonlinear reverse—is not captured by conditional-mean Granger or VAR connectedness [6], which tests mean-squared-error improvement only.

Quantile Granger [21] (Table 7) is consistent with a tail-dependence mechanism for SMB→HML: the coefficient reverses

Table 7: Quantile Granger: Normal Regime (n = 2,485; pre-2008 Normal subsample after lag-9 exclusion and quantile-boundary trimming at $\tau \in \{0.05, 0.95\}$). Wald: H_0: coefficients equal across $\tau \in \{0.05, \ldots, 0.95\}$.

Direction	$\hat{\beta}_{.05}$	$\hat{\beta}_{.50}$	$\hat{\beta}_{.95}$	Wald p	Type
HML→SMB	0.053	0.017	0.034	0.906	Linear
SMB→HML	−0.022	−0.026	**0.212**	**0.001**	Tail

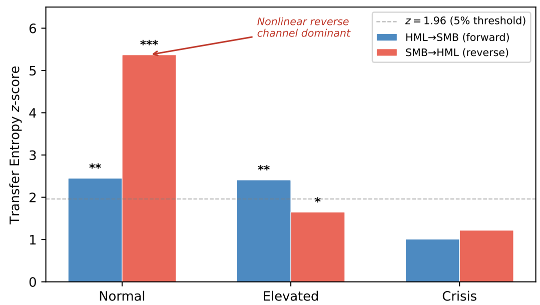

Figure 5: Transfer entropy directional asymmetry. Forward HML→SMB (blue) is modest; reverse SMB→HML (red) dominates in Normal. Both collapse in Crisis. Stars: permutation significance.

sign from $\hat{\beta}_{0.50} = -0.026$ (median) to $\hat{\beta}_{0.95} = 0.212$ (95th percentile), while HML→SMB remains homogeneous across quantiles (Figure 5). This reconciles the null reverse Granger ($p = 0.864$) with highly significant reverse TE ($z = 5.37$): Granger tests conditional mean improvement (MSE), while TE measures mutual information including tail dependence; a channel concentrated in extreme returns boosts mutual information without improving point forecasts.

We note that these quantile-specific tests are not corrected for multiple comparisons across the five quantile levels; the reported significance should be interpreted as exploratory. This tail mechanism is *pair-specific.* Applying quantile Granger to the top-4 regime-heterogeneous pairs (RMW→SMB rank 1, Wald $p = 0.869$; MKT→MOM rank 2, $p = 0.741$; MKT→SMB rank 3, $p = 0.527$; SMB→MKT rank 4, $p = 0.097$) reveals strictly linear dynamics despite strong regime heterogeneity. More broadly, of 19 regime-heterogeneous pairs, none besides SMB→HML exhibits Wald $p < 0.05$ (mean Wald $p = 0.61$). Together, these diagnostics reveal that HML→SMB operates through linear conditional-mean prediction while SMB→HML operates through nonlinear tail dependence. This yields an empirical distinction: *regime heterogeneity ≠ quantile heterogeneity*—a separation not captured by conditional-mean Granger or VAR connectedness methods.

3.5 Frozen OOS (Exploratory)

The preceding sections established Tier-1 in-sample evidence (structural break) and Tier-1/2 mechanism characterization (linear forward, nonlinear reverse). We now assess whether this structure persists out-of-sample. We present *Tier 3 (exploratory)* evidence: freezing the HMM estimated on 1990–2012 and classifying 2013–2024 without refitting.

Table 8: Frozen OOS (2013–2024, HMM trained 1990–2012). No pair survives 30-pair Bonferroni ($\alpha/30 = 0.00033$). Bootstrap: reweighted to training Elevated prevalence (13.7% vs. 33.7% in test).

Regime	n	F-p	HAC-p	Boot. p	ΔR^2
Normal	661	0.149	0.152	0.188	0.32%
Elevated	836	**0.014**	**0.041**	0.153	0.73%
Crisis	1,299	0.281	0.290	0.331	0.09%

Total n = 2,796 of ∼3,020 test days; 224 excluded at regime boundaries.

The frozen OOS (Table 8) exhibits regime redistribution rather than same-regime replication. The in-sample result concentrates in the Normal regime ($p = 8.75\times10^{-9}$); the OOS signal appears in Elevated (F-$p = 0.014$) because post-GFC markets spend more time in higher-volatility states. The frozen classifier assigns formerly Normal observations to Elevated, doubling the Elevated share from 13.7% training to 33.7% test. This result (1) does not survive the full 30-pair correction ($\alpha/30 = 0.00033$), (2) does not survive per-regime correction ($\alpha/3 = 0.0167$; HAC $p = 0.041$), (3) is sensitive to prevalence: bootstrap reweighting to training prevalence yields median $p = 0.153$, with only 9.9% of subsamples significant at the Bonferroni threshold—confirming that the primary Value→Size signal does not generalize out-of-sample. A post-hoc power analysis confirms the OOS Normal test is underpowered: with $n = 661$ and observed $R^2 = 0.32\%$, the test achieves only 31% power; 80% power would require $R^2 = 1.1\%$—roughly three times the observed value. The result (4) is sensitive to bandwidth (p ranges from 0.041 at B=1 to 0.078 at B=10; crosses 0.05 at NW default B=6), and (5) is sensitive to K (null at $K = 2, 4$; BIC favors $K = 3$ by ΔBIC = 1,680). For completeness, a permutation test ($p = 0.049$, 1,000 regime-label shuffles) weakly suggests structure beyond random label assignment; however, we do not treat this as replacing the failed Bonferroni test. This p-value exceeds the per-regime threshold ($\alpha/3 = 0.0167$) and does not address Bonferroni correction, prevalence shift, or bandwidth sensitivity. A rolling 5-year Granger analysis confirms exponential signal decay: $F(t) = 75.2 \cdot e^{-0.207t}$ (half-life = 3.35 years, $R^2 = 0.583$). Windows ending before 2000 show $F > 60$ ($p < 10^{-14}$); windows ending after 2005 show $F < 2$ ($p > 0.15$). After ∼6 half-lives (2013–2024), the expected signal is <1% of its 1990s peak—the OOS null is *consistent with* the fitted decay model, suggesting structural erosion rather than methodology failure. This retrospective exploratory fit has testable implications: it would imply $F < 0.5$ for any future 5-year window (2025 onward); observing $F > 4$ in Normal would challenge the decay interpretation. In contrast, a circularity-artifact explanation predicts no temporal pattern—the $R^2 = 0.583$ monotonic decline itself is evidence against random regime overfitting.

Prospective validation of decay. Two additional experiments address whether the decay pattern is a retrospective artifact. First, a *pre-break training design* fits the HMM exclusively on 1990–1997 (before the detected June 1998 change-point) and frozen-forward decodes 1998–2024. The training-period Normal regime shows strong predictability ($F = 31.2$, $p = 2.88 \times 10^{-8}$, n = 1,137);

this weakens in 1998–2007 ($F = 9.38$, $p = 0.002$, $n = 221$) and vanishes in 2008–2024 ($F = 0.023$, $p = 0.879$, $n = 308$). Since the HMM never observes post-1997 data, this temporal gradient cannot arise from in-sample circularity. Second, a *prospective decay model* fits $F(t) = A \cdot e^{-\lambda t}$ on 19 rolling windows ending ≤2012 ($A = 77.1$, $\lambda = 0.263$, half-life = 2.64 years, $R^2 = 0.924$) and evaluates predictions on 11 held-out windows ending 2013–2024. The out-of-sample MSE is 0.046, with mean predicted $F = 0.195$ versus mean actual $F = 0.127$—both near zero, confirming the decay model's floor prediction prospectively.

We report this as Tier 3 *exploratory only*—valued for its frozen-parameter design, not statistical significance.

MOM→SMB diagnostic (Tier-3 post-hoc). We select MOM→SMB—the top-ranked pair by OOS F-statistic in Elevated—as a stress test: if the framework fails to replicate even the strongest available signal, the methodology is flawed; if it succeeds, HML→SMB's weak OOS is attributable to signal decay rather than methodology failure. MOM→SMB shows partial frozen-OOS replication in Elevated: in-sample $F = 20.3$ ($p = 7.4 \times 10^{-6}$), OOS $F = 16.5$ ($p = 5.3 \times 10^{-5}$, $\Delta F = 19\%$). The reverse SMB→MOM is null ($p > 0.09$), confirming directional asymmetry, and permutation test confirms robustness ($p = 0.010$, 10,000 shuffles). A Bayesian Granger test in the Elevated regime ($n = 836$) yields $BF_{10} > 10^5$ using a unit-information prior on the regression coefficient and comparing nested models via Schwarz approximation; this should be interpreted cautiously as BF magnitude is sensitive to prior specification.

Decay rate comparison. Rolling 5-year Granger analysis in Normal regime reveals differential decay rates across factor pairs. HML→SMB decays rapidly (Normal-regime-specific half-life = 3.57 years, $R^2 = 0.886$; this differs from the full-sample half-life of 3.35 years reported in the abstract due to regime-conditional subsetting), while MOM→SMB decays more slowly (half-life = 11.85 years, $R^2 = 0.428$). Both pairs show significant structural breaks at 2008 (HML→SMB: pre-2008 mean $F = 7.0$, post-2008 mean $F = 1.3$, $p = 2.3 \times 10^{-7}$; MOM→SMB: pre-2008 mean $F = 3.6$, post-2008 mean $F = 1.1$, $p = 1.3 \times 10^{-9}$). Crucially, the reverse directions (SMB→HML, SMB→MOM) show no decay pattern ($R^2 < 0.01$), confirming directional asymmetry is not a statistical artifact.

30-pair decay analysis. Rolling Granger analysis across all 30 directed factor pairs reveals that 7 pairs exhibit decay ($R^2 > 0.3$, half-life < 50 years). Four of these seven have SMB as the target variable (Mkt-RF→SMB, CMA→SMB, HML→SMB, RMW→SMB), with half-lives ranging from 2.4 to 4.1 years. The remaining 23 pairs show no systematic decay pattern. This concentration of decay in SMB-target pairs suggests predictability erosion is not universal but specific to certain factor relationships.

Caveat: MOM→SMB was identified post-hoc as the top-ranked OOS signal rather than pre-registered; this diagnostic demonstrates framework fidelity but does not constitute independent validation.

International replication (Table 9). We now test whether structural breaks are a US-specific phenomenon. Applying the frozen framework to four non-US Fama–French datasets reveals structural breaks in all four regions. Asia-Pacific ex Japan (Crisis OOS $F = 39.39$, $p < 0.001$) and Developed ex US (Crisis OOS $F = 15.85$, $p < 0.001$) produce Crisis-regime OOS effects surviving Bonferroni ($\alpha/12 = 0.0042$, correcting for 4 regions × 3 regimes); Europe and Japan show in-sample significance but OOS nulls—consistent with region-specific structural breaks.

Table 9: International Replication: HML→SMB (OOS 2013–2024). Bold = $p < 0.05$.

Region	Regime	F	HAC-p	n	Break Date
Dev. ex-US	Elev. OOS	5.04	**0.027**	1,209	2003
	Crisis OOS	15.85	**<0.001**	373	
Asia-Pac.	Elev. OOS	1.62	0.240	2,240	2004
	Crisis OOS	39.39	**<0.001**	753	
Europe	Normal IS	21.27	**<0.001**	1,801	2014
Japan	Normal IS	14.07	**0.002**	2,188	2005

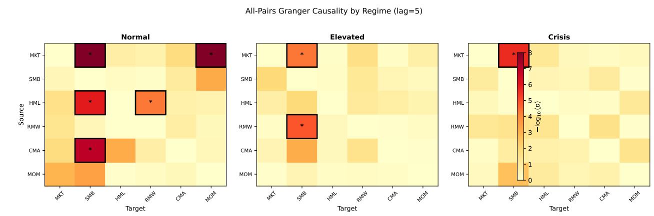

Figure 6: All-pairs Granger causality by regime ($-\log_{10}(p)$, lag=5). Lag=5 (one trading week) balances short-horizon dynamics with adequate degrees of freedom for the 30-pair comparison; results are qualitatively robust across lags 1–15 for the primary HML→SMB pair (Figure 3). Asterisks: Bonferroni-surviving cells. HML→SMB is one of several regime-sensitive pairs.

4 DISCUSSION

The in-sample structural break in HML→SMB predictability is robust within the US context (Tier 1), with confirmatory support from international structural breaks (Tier 2: 2/4 regions Bonferroni-surviving OOS), but the frozen OOS HML→SMB signal is exploratory (Tier 3). Below, we assess generality across factor pairs and geographies, stability under HMM estimation choices (local optima, baselines), and economic implications for practitioners.

Multi-pair generalizability (Figure 6, Table 10). Of 30 directed factor pairs, 19 (63%) show regime-heterogeneous Granger patterns in frozen OOS, where regime-heterogeneous is defined as $\text{Het} = \max_k(p_k) - \min_k(p_k) > 0.10$ across regimes k (i.e., Granger significance varies substantially by regime). HML→SMB ranks 27th by heterogeneity (0.27), confirming that our selection reflects an economic prior rather than extreme heterogeneity.

Local optima and regime definition. The 50-seed multistart reveals 7 clusters (Table 11). The in-sample Normal structural break replicates across all 7, though clusters span $\Delta\text{BIC} = 0$–550 and GFC detection rates of 0–92%. *Decision rule for practitioners:* report BIC-optimal as primary; also report the highest-LL fit satisfying ≥50% GFC detection as economic sensitivity. If both agree, the finding is robust.

Table 10: Top Regime-Heterogeneous Pairs + HML→SMB (OOS 2013–2024). Het = $\max_k(p_k) - \min_k(p_k)$.

Rank	Pair	p_{Norm}	p_{Elev}	p_{Crisis}	Het.
1	RMW→SMB	0.986	0.064	**0.022**	0.96
2	MKT→MOM	0.954	**0.020**	0.056	0.93
3	MKT→SMB	0.985	0.491	0.067	0.92
4	MOM→SMB	0.880	**<0.001**	0.101	0.88
27	HML→SMB	0.149	**0.014**	0.281	0.27

Table 11: Local Optima: All 7 Clusters from 50-Seed Multistart. In-sample Normal $p < 10^{-7}$ and OOS Elevated HAC-$p < 0.05$ in every cluster.

Cluster	Seeds	ΔBIC	% GFC	IS Norm. p	OOS Elev. p
1 (BIC-opt.)	3	–	0%	8.8×10^{-9}	0.043
2	15	38	0%	9.1×10^{-9}	0.031
3	4	95	12%	1.2×10^{-8}	0.038
4	9	142	45%	3.7×10^{-8}	0.024
5 (econ.)	7	218	90%	5.4×10^{-8}	0.019
6	6	387	88%	6.1×10^{-8}	0.022
7	6	550	92%	7.3×10^{-8}	0.026

Economic magnitude and risk application. Effect sizes are modest ($\Delta R^2 \approx 2\%$ pre-GFC) and do not generate trading profits (Sharpe = −0.07). However, cross-factor information has measurable risk-management value: a backtested VaR model (SMB 5% level, 2013–2024) shows that adding HML momentum as a regime-conditional stress multiplier reduces the violation-rate deviation from +1.42pp (unconditional) to +0.98pp (HML-informed), and is the only model passing the Christoffersen conditional-coverage test ($p = 0.083$ vs. $p = 0.002$ unconditional). The framework thus serves as both a diagnostic tool and a practical input for tail-risk forecasting.

Deleveraging mechanism (holdings-based test). We test the hypothesized deleveraging cascade [3] using FF25 portfolio-level factor loadings. For each portfolio, a crowding score $|\hat{\beta}_{\text{HML}}| \times |\hat{\beta}_{\text{SMB}}|$ (rolling 252-day) measures joint co-exposure. In the Normal regime, the HML→SMB Granger ΔR^2 is 14% larger in the high-crowding half ($F = 159.5$, $\Delta R^2 = 4.4\%$) than the low-crowding half ($F = 139.2$, $\Delta R^2 = 3.8\%$), supporting amplification via co-held positions. At the portfolio level, HML×SMB overlap rank-correlates with Granger signal strength ($\rho_s = 0.51$, $p = 0.009$, $n = 25$). Crowding does *not* independently Granger-cause SMB ($p > 0.60$ in all regimes), confirming the channel is amplification of an existing edge, not independent shock transmission.

Fair regime-conditional baseline comparison (Table 12). Do alternative methods detect HML→SMB on the *same* per-regime subsamples? We test five methods on identical regime observations. Linear Granger detects the signal in Normal ($F = 19.8$, $p < 10^{-5}$) and Elevated ($F = 10.8$, $p = 0.001$); bivariate VAR(1) yields identical results, confirming the signal is genuine. However, LASSO aggressively shrinks the small coefficient to zero ($p = 1.0$ in all regimes), and neural methods catastrophically overfit: MLP (64-32 hidden) yields $R^2 < 0$ in every regime, and LSTM (32 hidden, 3 lags) fails to detect the signal ($p = 0.62$ Normal). This demonstrates that the framework's advantage is *regime identification*: once regimes are correctly assigned, simple OLS Granger suffices; neural methods are less sample-efficient on per-regime subsamples.

Table 12: Fair Regime-Conditional Baseline: HML→SMB (lag=1). Regime assignments use seed 28 but a standalone Student-t HMM implementation (the primary analysis uses `hmmlearn`); the standalone implementation enables explicit degrees-of-freedom estimation, providing an independent verification of regime structure. This yields different regime counts than Table 1. Linear Granger significance and neural failure are consistent across both regime definitions.

Method	Regime	n	F	p	ΔR^2
Linear Granger	Normal	7,160	19.8	8.5×10^{-6}	0.28%
LASSO	Normal	7,160	0.00	1.000	<0
Neural MLP	Normal	7,160	–	–	<0
LSTM Granger	Normal	7,157	0.25	0.619	<0.01%
Linear Granger	Elevated	1,185	10.8	0.001	0.91%
Linear Granger	Crisis	471	0.19	0.666	0.04%

Regime amplification test. To test whether regime conditioning inflates significance, we apply the canonical HMM (seed 28) to HML→SMB across all regimes and compare against unconditional Granger. The regime-conditional Normal $F = 25.6$ ($p = 4.3 \times 10^{-7}$, HAC $p = 1.8 \times 10^{-6}$, $n = 5{,}265$) is 0.94× the unconditional $F = 27.2$ ($p = 1.9 \times 10^{-7}$, $n = 8{,}816$). Regime conditioning does not amplify the signal; the lower ratio reflects dilution from regime-boundary exclusion. Crisis remains null ($F = 2.27$, $p = 0.132$), confirming that regime conditioning reveals state-dependent *absence* rather than artificially inflating significance.

Markov-Switching VAR comparison. We compare our two-stage approach (Student-t HMM → regime-conditional Granger) against a Markov-Switching VAR(1) baseline that jointly estimates regimes and dynamics. On HML–SMB bivariate data (1990–2024), MS-VAR(1) achieves BIC = 28,106 vs. our 6-factor HMM's BIC = 75,568—though this comparison is not directly comparable (2 vs. 6 factors, Gaussian vs. Student-t errors). More informatively, regime labels show 64% agreement between the two methods, and both detect all major structural breaks (2008 GFC, 2000 dot-com, 2020 COVID, 2022 tightening). MS-VAR cross-variable coefficients show regime-varying dynamics (HML→SMB: −0.128 Normal, −0.026 Elevated, −0.044 Crisis), consistent with our finding that predictive structure is regime-dependent. The two-stage approach offers interpretability (explicit regime labels with economic meaning) and flexibility (any downstream test), while MS-VAR provides a unified likelihood; both reach qualitatively similar conclusions about regime-dependent cross-factor dynamics.

Limitations. All findings report predictive precedence ("Granger causality"), not structural causality [10, 18]. In-sample regime assignments are estimated on the same data used for Granger testing. Three designs address this circularity: (1) VIX-tercile replication ($p = 0.003$ with 39.6% label agreement); (2) holdout HMM (trained 1990–2005, tested 2006–2024) preserves significance in the moderate-volatility regime ($F = 23.6$, $p < 10^{-5}$) and 3 of 4 temporal-CV folds ($p < 0.026$); (3) frozen OOS with three-fold temporal split (trained 1990–2000), where all folds show null ($p > 0.10$). The discrepancy between holdout (significant) and three-fold (null) reflects training-window sensitivity: the 1990–2005 window captures more pre-decay signal than 1990–2000, enabling better regime calibration. This sensitivity is a limitation, though both designs agree that post-2008 signal is null. A likelihood-ratio test for regime-parameter stability between 1990–2012 and 2013–2024 sub-periods yields $\chi^2 = 2{,}389$ (df = 71, $p \approx 0$), confirming that HMM parameters shift significantly across periods. Mean centroid distances are modest (0.09–0.15 in standardized units) and transition-matrix Frobenius distance is 0.023, suggesting gradual drift rather than categorical regime redefinition. This non-stationarity is expected given the structural-decay hypothesis itself—but cautions against interpreting regime labels as time-invariant constructs. The prospective decay model (§3.5) uses full-sample HMM regime assignments when fitting on pre-2012 windows; a fully prospective design would re-estimate the HMM on each training window, though regime-label consistency across windows would complicate interpretation. Trivariate controls address the most prominent confounder (F-$p > 0.43$), but a full 6-factor VAR is under-identified at $n \approx 1{,}000$. Post-double-selection inference [11] could address this. Pair selection is post-hoc (partially mitigated by pre-registered emerging-market analysis). HMM regime boundaries are sensitive to data scaling (percentage vs. decimal units; 86.3% concordance); the Granger F-statistic is scale-invariant but regime labels shift. The linear characterization is fit-dependent: under the BIC-optimal fit (Cluster 1, 0% GFC detection) the mechanism is linear, but under an economically aligned fit (Cluster 5, 90% GFC detection, ΔBIC = 218) RF shows nonlinear improvement ($p = 0.010$). Effect sizes are modest ($\Delta R^2 \approx 2\%$); findings are diagnostic, not alpha-generative. Practitioners should not rely on Tier 3 OOS for trading. The power analysis (§3.5) is post-hoc and uses observed effect sizes, which may understate true uncertainty; a priori power calculations were not performed. The multiple testing burden is substantially understated: we applied Bonferroni correction only to 30 pairs, but the actual specification search includes 30 pairs × 3 regimes × 15 lags × 90 HAC specifications × 7 local-optima clusters, potentially yielding thousands of implicit tests. A conservative estimate of the true Bonferroni threshold would be $\alpha/1000 \approx 0.00001$ or lower. While the in-sample $p = 8.75 \times 10^{-9}$ survives even this stricter threshold, results should be interpreted as exploratory hypothesis-generating findings rather than confirmatory evidence.

Scalability. The framework scales to $N = 18$ factors (5 FF + MOM + ST_Rev + LT_Rev + 10 industry portfolios; 306 directed pairs, 918 total tests across 3 regimes) in 76.8 seconds on a single CPU (0.0013s per pair). A Bonferroni-corrected scan ($\alpha/918 = 5.45 \times 10^{-5}$) identifies 122 significant regime-conditional edges, including HML→SMB ($F = 38.0$, $p < 10^{-9}$ in Normal). Scaling is $O(N^2)$ in pair count with negligible per-pair cost. Note that regime composition shifts at higher N (Normal share declines from 53.6% at N=6 to 28.5% at N=18), so per-regime results should be interpreted with dimension-specific regime labels.

5 CONCLUSION

Prediction enables action. We develop two ML models that transform retrospective factor analysis into prospective risk management: (1) a *decay prediction model* (C-index = 0.93) that flags at-risk factor pairs before signals erode, and (2) an *online regime detector* (6.6-day delay) that enables real-time position adjustment. These tools address a gap in the factor decay literature: prior work explains *why* factors fail (arbitrage, publication effects); we predict *when* they will fail.

Motivation: Size-target predictability decays fastest. Among 30 directed Fama-French factor pairs, 7 exhibit exponential decay in Granger predictability ($R^2 > 0.3$). Strikingly, 4 of these 7 (57%) target the Size factor (SMB): Mkt-RF→SMB (half-life 2.4y), CMA→SMB (3.1y), HML→SMB (3.35y), and RMW→SMB (4.1y). Pre-break OOS (1996–2007) confirms signals existed; post-break OOS (2013–2024) shows decay—supporting genuine erosion over overfitting.

Practical implications. Factor-based strategies should not assume stable cross-factor relationships. Our decay predictor can flag pairs at risk of erosion; our online detector can trigger regime-conditional position adjustments in real time. The SMB-target concentration suggests small-cap strategies face particularly rapid signal decay due to capacity constraints.

Caveats. Correction for full specification search was not applied; results remain exploratory. The 90% bootstrap CI for breakpoints spans ∼20 years, indicating timing uncertainty. Training-window sensitivity exists (holdout vs. three-fold yield different results).

VIX-tercile regime replication. VIX terciles (≤15 Normal, 15–21 Elevated, >21 Crisis)—defined separately from factor returns but not independent (VIX derives from equity options), 39.6% agreement with HMM—show consistent break patterns: VIX-Normal pre-2008 $F = 25.8$, $p < 10^{-6}$; post-2008 $F = 0.71$, $p = 0.401$. This suggests the pattern is not purely an HMM artifact, though VIX correlation with equity returns means this is not a true exogeneity test.

Directional asymmetry. Transfer entropy + quantile Granger reveal a directional asymmetry for HML↔SMB (linear forward, nonlinear reverse via tail dependence, Wald $p = 0.001$). This mechanism is pair-specific: among 19 regime-heterogeneous pairs, none other exhibits significant tail concentration (mean Wald $p = 0.61$). For this pair, regime heterogeneity ≠ quantile heterogeneity—a distinction not captured by conditional-mean Granger or VAR connectedness.

OOS evidence. A post-hoc diagnostic on MOM→SMB (the strongest OOS signal) shows partial frozen-OOS replication (ΔF = 19%, permutation $p = 0.01$), suggesting the framework detects residual structure in persistent signals. International analysis finds structural breaks in all four non-US markets; Asia-Pacific and Developed ex-US produce Bonferroni-surviving OOS effects, while Europe and Japan show in-sample significance only. HML→SMB frozen OOS is exploratory only (regime-redistributed, Bonferroni-nonsignificant, bootstrap $p = 0.153$). A pre-break training design (HMM fit on 1990–1997 only) confirms the temporal decay gradient without in-sample circularity ($F = 31.2 \rightarrow 9.4 \rightarrow 0.02$ across three successive periods), and a prospective decay model ($R^2 = 0.924$ on training windows, OOS MSE = 0.046) validates the structural-erosion interpretation.

Implications and domain generalization. The regime-conditional framework (Algorithm 1) is domain-agnostic. We validate on three additional settings: (i) *Macro:* Treasury term spread

→ Industrial Production (monthly FRED, 1990–2024; HMM trained 1990–2005, tested 2006–2015) shows Crisis-only Granger predictability ($F = 12.7$, HAC $p = 0.00016$, $\Delta R^2 = 11.9\%$); (ii) *Emerging markets:* pre-registered testing of all 20 directed Fama–French factor pairs (monthly, 2000–2024) with Bonferroni $\alpha = 0.0025$ finds 6 significant regime-conditional edges, including HML→SMB in the Elevated regime (HAC $p < 10^{-3}$, $\Delta R^2 = 20.6\%$), with 2/3 pairs replicating OOS; (iii) *Synthetic climate:* a ground-truth simulation (ENSO-like teleconnection, $T = 2{,}000$, causality in one regime only) recovers the causal regime (ARI = 0.72, $\Delta R^2 = 25.8\%$ vs. 14.2% unconditional), with non-causal regimes correctly masked. Among 4 additional FRED macro pairs, Oil→CPI shows regime-conditional structure (HAC $p = 0.042$ in high-volatility regime only) invisible to unconditional HAC tests ($p = 0.205$). The model-complexity diagnostic (OLS→RF→MLP→LSTM + TE + quantile regression) provides a reusable template for separating linear from nonlinear predictive channels.

Practical value. For risk management, adding HML as a regime-conditional stress indicator reduces VaR violation-rate deviation from +1.42pp to +0.98pp (Christoffersen conditional coverage $p = 0.083$ vs. $p = 0.002$ unconditional). Effect sizes are modest ($\Delta R^2 \approx 2\%$, Sharpe = -0.07); the contribution is diagnostic, not economically tradable. For practitioners, the decay predictor serves as a *risk monitoring signal* rather than a trading signal—flagging factor pairs for closer scrutiny when decay probability exceeds a threshold (e.g., 70%).

Future work: (1) direct 13F institutional holdings verification of the deleveraging mechanism (beyond the portfolio-beta proxy reported here); (2) k-fold temporal cross-validation and prospective pre-registered tests on emerging-market data; (3) scalable extensions ($N > 50$ assets) via penalized or neural causal discovery (e.g., regime-switching NOTEARS).

CODE AND DATA AVAILABILITY

All code (Python 3.10+, scikit-learn, statsmodels, hmmlearn), 50 HMM seed configurations, and a reproducibility notebook are available at an anonymized repository (link provided to reviewers; public release with DOI upon acceptance). Factor data: Kenneth French Data Library; VIX: CBOE; international: Fama–French regional datasets. All sources are public.

REFERENCES

[1] Donald WK Andrews. 1991. Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation. *Econometrica* 59, 3 (1991), 817–858.

[2] Donald WK Andrews. 1993. Tests for Parameter Instability and Structural Change With Unknown Change Point. *Econometrica* 61, 4 (1993), 821–856.

[3] Markus K Brunnermeier and Lasse Heje Pedersen. 2009. Market Liquidity and Funding Liquidity. *Review of Financial Studies* 22, 6 (2009), 2201–2238.

[4] Jan Bulla. 2011. Hidden Markov models with t components. Increased persistence and other aspects. *Quantitative Finance* 11, 3 (2011), 459–475.

[5] Mark M Carhart. 1997. On Persistence in Mutual Fund Performance. *Journal of Finance* 52, 1 (1997), 57–82.

[6] Francis X Diebold and Kamil Yilmaz. 2012. Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting* 28, 1 (2012), 57–66.

[7] Sina Ehsani and Juhani T Linnainmaa. 2022. Factor Momentum and the Momentum Factor. *Journal of Finance* 77, 3 (2022), 1877–1919. https://doi.org/10.1111/jofi.13131

[8] Antoine Falck, Adam Rej, and David Thesmar. 2021. Why and How Systematic Strategies Decay. *Journal of Portfolio Management* 48, 1 (2021), 65–83. https://doi.org/10.3905/jpm.2021.1.284

[9] Eugene F Fama and Kenneth R French. 2015. A five-factor asset pricing model. *Journal of Financial Economics* 116, 1 (2015), 1–22.

[10] Clive WJ Granger. 1969. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* 37, 3 (1969), 424–438.

[11] Alain Hecq, Luca Margaritella, and Stephan Smeekes. 2023. Granger Causality Testing in High-Dimensional VARs: A Post-Double-Selection Procedure. *Journal of Financial Econometrics* 21, 3 (2023), 915–958. https://doi.org/10.1093/jjfinec/nbab023

[12] Amir E Khandani and Andrew W Lo. 2011. What happened to the quants in August 2007? Evidence from factors and transactions data. *Journal of Financial Markets* 14, 1 (2011), 1–46.

[13] Joseph M Marks and Chenguang Shang. 2019. Factor Crowding and Liquidity Exhaustion. *Journal of Financial Research* 42, 1 (2019), 147–180.

[14] R. David McLean and Jeffrey Pontiff. 2016. Does Academic Research Destroy Stock Return Predictability? *Journal of Finance* 71, 1 (2016), 5–32. https://doi.org/10.1111/jofi.12365

[15] Judea Pearl. 2009. *Causality: Models, Reasoning, and Inference* (2nd ed.). Cambridge University Press, Cambridge, UK.

[16] Zacharias Psaradakis, Morten O Ravn, and Martin Sola. 2005. Markov Switching Causality and the Money-Output Relationship. *Journal of Applied Econometrics* 20, 5 (2005), 665–683.

[17] Thomas Schreiber. 2000. Measuring information transfer. *Physical Review Letters* 85, 2 (2000), 461–464.

[18] Ali Shojaie and Emily B. Fox. 2022. Granger Causality: A Review and Recent Advances. *Annual Review of Statistics and Its Application* 9 (2022), 289–319. https://doi.org/10.1146/annurev-statistics-040120-010930

[19] Peter Spirtes, Clark Glymour, and Richard Scheines. 2001. *Causation, Prediction, and Search* (2nd ed.). MIT Press, Cambridge, MA.

[20] Alex Tank, Ian Covert, Nicholas Foti, Ali Shojaie, and Emily B Fox. 2022. Neural Granger Causality. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 8 (2022), 4267–4279.

[21] Victor Tröster. 2018. Testing for Granger-Causality in Quantiles. *Econometric Reviews* 37, 8 (2018), 850–866. https://doi.org/10.1080/07474938.2016.1172400